Research Project Periodic Log Entry Form

01st -15th April

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
1 April	Participated in a group meeting and discussed how the labor management component will integrate with the overall system.
3 April	Studied labor management practices in tea estates and identified key challenges such as worker allocation and attendance tracking.
4 April	Reviewed several research papers and online resources related to workforce management systems.
6 April	Identified key functional requirements including scheduling, attendance tracking, and worker performance monitoring.
8 April	Explored possible technologies and tools for developing the module such as web dashboards and database systems.
10 April	Started drafting initial system design ideas and discussed them with the group for further improvements.

Supervisor's Comments

- Student progressing well in research project : YES / NO

  If NO

  Project work to be carried out before next evaluation period:
- Code base checked : YES / NO

Signature : ..... Date : 20/04/2026

Research Project Periodic Log Entry Form

15th-30 April

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
16 April	Defined worker health monitoring parameters (heart rate, SpO2, and derived features) for risk prediction model.
17 April	Prepared dataset and performed preprocessing including feature selection and data cleaning for model training.
19-20 April	Trained machine learning model to predict worker health risk levels based on collected features.
21-23 April	Evaluated model performance and generated outputs such as risk level, confidence, and health score.
24-26 April	Deployed trained model using Hugging Face Spaces and implemented backend using Python (Gradio interface).
27 April	Developed user interface for inputting worker data and displaying prediction results including risk level and recommendations.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : ..... Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th May

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
01 May	Researched tea estate labor management requirements and worker allocation process.
03 May	Designed labor task management workflow.
06 May	Developed task allocation logic and worker tracking structure.
09 May	Created backend database structure for labor records.
12 May	Demonstrated labor management prototype during PP1 presentation.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th May

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
16 May	Improved worker attendance and task allocation module functionalities.
19 May	Added labor performance tracking and productivity calculation features.
22 May	Enhanced labor management dashboard and reporting interface.
26 May	Integrated labor records with centralized project database.
29 May	Conducted testing and fixed workflow issues in labor allocation system.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th June

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
01 June	Enhanced worker attendance tracking and labor record management.
03 June	Implemented worker productivity analysis and performance metrics.
07 June	Added welfare monitoring features and worker statistics reporting.
10 June	Integrated labor management dashboard with centralized database.
13 June	Performed testing and resolved labor allocation workflow issues.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th June

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
16 June	Improved worker welfare monitoring functionalities based on testing feedback.
19 June	Enhanced labor performance reporting and dashboard visualization.
22 June	Integrated the Labor Management & Welfare module with other project modules.
26 June	Optimized worker productivity analysis and reporting functionalities.
29 June	Conducted integration testing and resolved identified implementation issues.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th July

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
02 July	Refined worker welfare monitoring functionalities to improve system usability.
05 July	Enhanced labor performance reporting with additional analytical insights.
08 July	Improved dashboard visualization for worker productivity and welfare information.
11 July	Conducted validation testing of labor management functionalities.
14 July	Resolved identified issues and optimized overall module performance.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th July

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
16 July	Enhanced worker productivity analysis using refined performance indicators.
19 July	Improved labor management reporting and dashboard responsiveness.
22 July	Conducted module integration testing with the centralized management system.
26 July	Optimized data processing and reporting functionalities based on testing outcomes.
29 July	Updated implementation documentation and prepared the module for further evaluation.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th August

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
02 August	Refined labor management functionalities based on validation testing feedback.
05 August	Enhanced worker welfare monitoring and performance reporting features.
08 August	Improved dashboard visualization and reporting accuracy.
11 August	Conducted module validation testing and resolved identified issues.
14 August	Updated implementation documentation with recent module improvements.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th August

Project ID : R26-IT-118

Student ID : IT22267986

Name : Gunasekara L.M.N.P

Individual Component : Labor Management & Welfare Optimization

Summary of RP work carried out during the reporting period

Date	RP work
16 August	Enhanced the Labor Management & Welfare module based on recent implementation feedback.
19 August	Improved integration between labor management functionalities and other project modules.
22 August t	Conducted functional and integration testing of the Labor Management & Welfare module.
26 August	Updated presentation materials and technical documentation for Progress Presentation II (PP2).
28 August	Prepared the IoT health monitoring module and supporting materials for Progress Presentation II (PP2).

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026